Feynman's Path Integral formulation: the culmination of the First Quantum Revolution

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We provide an overview of Feynman's *Space-Time Approach to Non-Relativistic Quantum Mechanics* [1], where he proposes a new formulation of Quantum Mechanics based on a least action principle, establishing an analogy with classical theories. After stating his postulates and main results, we show an application of Feynman's work by solving the harmonic oscillator and providing the main tools in order to work with perturbation theory. Finally, we discuss the pros and cons of his revolutionary formulation.

INTRODUCTION

In 1926, two different formulations of the quantum theory coexisted: the differential equation of Schrödinger [2] and the matrix algebra of Heisenberg [3]. Although they seemed discordant, they were proved to be mathematically equivalent one year later thanks to Dirac's transformation theory [4]. Unfortunately, both of them were founded on postulates that were born from different experiments. Because of that, they could not be related to the variational principle nor the classical action in a satisfactory way.

After years without major discoveries, Feynman achieved the formulation of Quantum Mechanics via a least action principle thanks to the Path Integral formulation, which was first postulated by Norbert Wiener [5] in 1922 to solve problems related to the Brownian motion of particles. Later, he took advantage of these new rules in order to formulate Quantum Electrodynamics and introduce the tool for which he is mainly known: Feynman diagrams [6].

This paper develops this non-relativistic formulation, which is mathematically analogous to the previous two postulated by Schrödinger and Heisenberg. Here, this new point of view establishes a clear relation with the classical theory thanks to the variational principle.

POSTULATES OF FEYNMAN'S FORMULATION

In order to establish the main differences between classical and quantum mechanics, let's imagine an experiment in which we perform three measurements, A, B and C, successive in time. If P_{ij} is the probability of measuring I and obtaining i, and then measuring J and getting j, the classical law of probabilities implies that

$$P_{ac} = \sum_b P_{ab} P_{bc}, \tag{1}$$

where we suppose that both events are independent. Surprisingly, in quantum mechanics this result is often false. Based on the work of Schrödinger [2], Born and Jordan [7] and Dirac [4], Feynman imposes the following law: *there exist complex numbers φ_{ab}, φ_{bc} and φ_{ac} such that*

$$P_{ab} = |\varphi_{ab}|^2,\ P_{bc} = |\varphi_{bc}|^2,\ P_{ac} = |\varphi_{ac}|^2; \tag{2}$$

so that Eq. (1) *is replaced by*

$$\varphi_{ac} = \sum_b \varphi_{ab}\varphi_{bc}. \tag{3}$$

Since Eq. (1) supposes that B has some definite value between measuring A and C, this must be necessarily false in quantum mechanics, as we would be taking into account some kind of attempt on measuring B. Eq. (3) implies that the probability of going from a to c is represented as the square of a sum of complex quantities, thus showing the phenomenon of interference. This represents the wave nature of matter, which is fundamental in the formulation of quantum mechanics.

This idea can be extended to define a probability amplitude for a completely specified space-time path. Suppose that we measure the 1-dimensional position $\{x_i\}$ of a particle at successive times $\{t_i\}$, where $t_{i+1} = t_i + \epsilon$, $\epsilon \ll 1$. These measurements are what Feynman defines as *ideal measurements*, where no more information than the position can be extracted without further disturbance to the system. If we define the region R as the set of possible ranges of $\{x_i\}$, Eqs. (2) and (3) tell us that the probability of finding the particle in the region R is $|\varphi(R)|^2$, where

$$\varphi(R) = \lim_{\epsilon \longrightarrow 0} \int_R \Phi(x_1, ..., x_N)\, \mathrm{d}x_1 \ldots \mathrm{d}x_N\,. \tag{4}$$

Here, the probability amplitude $\Phi(x_1, ..., x_N)$ is a function of the variables $\{x_i\}$ defining the path. Since $\epsilon \to 0$, this amplitude depends on the path $x(t)$.

With this definition in mind, we can set up the two following postulates [1]:

1. *If an ideal measurement is performed to determine whether a particle has a path lying in a region of space-time, then the probability that the result will be affirmative is the absolute square of a sum of complex contributions, one from each path in the region.*

2. *The paths contribute equally in magnitude, but the phase of each contribution is the classical action in units of $\hbar$, i.e. the time integral of the Lagrangian taken along the path.*

Whereas the first postulate is given by Eq. (4), the second one prescribes how to compute the probability amplitude $\Phi[x(t)]$ for a given path $x(t)$. In order to prove the mathematical equivalence with the other formulations, we will suppose from now on that the Lagrangian, $L(\dot{x}, x)$, is a quadratic function on the velocities.

To compute the action, $S = \int L(\dot{x}, x)\,\mathrm{d}t$, we need to know the path at all points, not only at $\{x_i\}$. We shall assume that the function $x(t) \in [x_i, x_{i+1}]$ is the path followed by the classical particle between t_i and t_{i+1}. Thus, since the classical path is the one which minimizes the action, we can write $S = \sum_i S(x_{i+1}, x_i)$, where

$$S(x_{i+1}, x_i) = \min \int_{t_i}^{t_{i+1}} L(\dot{x}(t), x(t))\,\mathrm{d}t\,. \tag{5}$$

In order to avoid divergences in the sum of S, we are going to consider a finite time interval, although this reflects a further incompleteness of the postulates. Combining these two, we find

$$\varphi(R) = \lim_{\epsilon \to 0} \int_R \exp\left\{\frac{i}{\hbar} \sum_i S(x_{i+1}, x_i)\right\} \frac{\mathrm{d}x_1}{A} \dots \frac{\mathrm{d}x_N}{A}, \tag{6}$$

where $1/A^N$ is a normalization factor introduced to avoid a divergence in the limit $\epsilon \to 0$.

DEFINITION OF THE WAVE FUNCTION

With the definition of $\varphi(R)$ in Eq. (6), the new formulation of quantum mechanics is completed. To prove that it is equivalent to the standard formulation, the quantum mechanical wave equation needs to be defined from what was previously stated.

To do this, it is important to divide the region R in the parts we are interested in measuring. Following this idea, it is safe to say that R can be divided into three separate sections for a particular time t:

- R' is a bounded region in space lying before a past time t', where $t' < t$.
- R'' is another restricted region that lies after a later time t'' such that $t'' > t$.
- The area that encapsulates all the unrestricted values of positions, x, will be the whole space-time lying between t' and t''.

Here, we only care about the bounded regions since the time regarding the unrestricted values of positions between t' and t'' can be considered arbitrarily small in relation to the other regions. Now, it is possible to redefine the probability of finding the path inside R, $|\varphi(R)|^2$, as the probability of the path being in R' before a past time t' and then in R'' after a future time t'', which will be denoted by $|\varphi(R', R'')|^2$. To relate this new regions to Eq. (6), we now say that the time t is now associated to a point k in this new time division with ϵ steps. Then, since the action is governed by $L(\dot{x}(t), x(t))$, the exponential on Eq. (6) can be rewritten as the product of two exponential functions that correspond to the paths on a previous and a later time than t:

$$\exp\left[\frac{i}{\hbar} \sum_{i=-\infty}^{k-1} S(x_{i+1}, x_i)\right] \times \exp\left[\frac{i}{\hbar} \sum_{i=k}^{\infty} S(x_{i+1}, x_i)\right]. \tag{7}$$

When integrating, both factors $i > k$ and $i < k$ will end up as a function of x_k. Integrating over x_k, we obtain a new definition for $\varphi(R', R'')$:

$$\varphi(R', R'') = \int \chi^*(x_k, t) \mathit{\Psi}(x_k, t)\,\mathrm{d}x_k\,, \tag{8}$$

where

$$\mathit{\Psi}(x_k, t) = \lim_{\epsilon \to 0} \int_{R'} \exp\left[\frac{i}{\hbar} \sum_{i=-\infty}^{k-1} S(x_{i+1}, x_i)\right] \frac{\mathrm{d}x_{k-1}}{A} \dots, \tag{9}$$

and

$$\chi^*(x_k, t) = \lim_{\epsilon \to 0} \int_{R''} \exp\left[\frac{i}{\hbar} \sum_{i=k}^{\infty} S(x_{i+1}, x_i)\right] \frac{1}{A} \frac{\mathrm{d}x_{k+1}}{A} \dots, \tag{10}$$

are functions that contain the information about the past and future history of the system, respectively. Also, we can note that they are independent to a change in one of the established states: if we were to change both the region and the Lagrangian for any of the established times, and we obtained the same $\mathit{\Psi}$ or χ^*, Eq. (8) would still hold, and we would find the same probability for the system to end in R'' if either R'' is changed to r'' or R' is changed to r'. With this in mind, we find a final definition for $\varphi(R', R'')$:

$$|\varphi(R', R'')|^2 = \left| \int \chi^*(x, t)\, \mathit{\Psi}(x, t)\ \mathrm{d}x \right|^2, \tag{11}$$

which is the probability that the system in state $\mathit{\Psi}$ will appear later in χ.

SCHRÖDINGER'S EQUATION

Now that we have obtained the wave equation for different sets of time, it is possible to derive Schrödinger's equation from what was previously stated. Here, we are going to use the case for the region R'. Since the equation for χ^* can be obtained in the same way, we start by looking at how Eq. (9) is computed at the next instant of time $\Psi(x_{k+1}, t+\epsilon)$. If one does so, it is found that there is only a factor $\frac{1}{A}\exp[iS(x_{k+1}, x_k)/\hbar]$ of difference w.r.t. Eq. (9). The integrations for this case on dx_i up to dx_{k-1} can be done with this factor omitted, being the result for those integrations $\Psi(x_k, t)$. Hence, we can write:

$$\Psi(x_{k+1}, t+\epsilon) = \int \exp\left[\frac{i}{\hbar} S(x_{k+1}, x_k)\right] \Psi(x_k, t) \frac{\mathrm{d}x_k}{A}. \tag{12}$$

We are going to apply Eq. (12) to the case of a particle in a one-dimensional motion with the presence of a potential $V(x)$. First, it is important to define some approximations for $S(x_{i+1}, x_i)$, since it is tedious to calculate its exact value for an arbitrary ϵ. For that, it is only necessary to have an approximation error of first or zero order in ϵ. Under the circumstances that the particle moves along a straight line, we can replace the integral $S(x_{i+1}, x_i)$ by the trapezoidal rule. If we consider no vector potential or other terms linear in the velocity we have:

$$S(x_{i+1}, x_i) = \epsilon L\left(\frac{x_{i+1} - x_i}{\epsilon}, x_{i+1}\right). \tag{13}$$

Then, for our particular case we can write:

$$S(x_{i+1}, x_i) = \frac{m\epsilon}{2}\left(\frac{x_{i+1} - x_i}{\epsilon}\right)^2 - \epsilon V(x_{i+1}). \tag{14}$$

Rewriting $x_{k+1} = x$ and $x_{k+1} - x_k = \xi$, which makes $x_k = x - \xi$, Eq. (12) becomes:

$$\Psi(x_{k+1}, t+\epsilon) = \tag{15}$$
$$\int \exp\left(\frac{im\xi^2}{2\hbar\epsilon}\right) \exp\left(\frac{-i\epsilon V(x)}{\hbar}\right) \Psi(x-\xi, t) \frac{\mathrm{d}\xi}{A}.$$

Here, ϵ can be taken arbitrarily small, which makes $\Psi(x-\xi, t)$ a smooth function in ξ. As ϵ is arbitrarilly small, the contribution of the factor $\exp\left(\frac{im}{\epsilon}\frac{\xi^2}{2\hbar}\right)$ becomes virtually negligible when ξ is big because the phase changes so rapidly that the positive and negative contributions cancel each other out. Also, because only small values of ξ are effective, we can expand $\Psi(x-\xi, t)$ around this variable. Then, we would see that $\int_{-\infty}^{\infty} \exp\bigl(im\xi^2/2\hbar\epsilon\bigr)\xi^n d\xi$ would vanish for odd n, but for even values it would give values -obtainable by applying Cauchy's residue theorem [8]- of really small order, with ξ^4 being at least of order ϵ. Then, if we expand the left side in Eq. (15) to first order in ϵ, set $A = (2\pi\hbar\epsilon i/m)^{\frac{1}{2}}$ so that both sides end up with zero order in ϵ, and expand the exponential containing $V(x)$, we end up with the following:

$$\Psi(x,t) + \epsilon\frac{\partial\Psi}{\partial t} = \left(1 - \frac{i\epsilon}{\hbar}V(x)\right) \times \left(\Psi(x,t) + \frac{\hbar\epsilon i}{2m}\frac{\partial^2\Psi}{\partial x^2}\right). \tag{16}$$

From this equation, cancelling $\Psi(x,t)$ from both sides, comparing terms to first order in ϵ and then multiplying by $i\hbar$, we can get Schrodinger's equation in its usual form:

$$-\frac{\hbar}{i}\frac{\partial\Psi}{\partial t} = \frac{1}{2m}\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 \Psi + V(x)\Psi. \tag{17}$$

CLASSICAL LIMIT

Let's now recap what this definition of the wave function exactly means. In classical mechanics, Huygens' principle states that each point at a known surface can be understood as a source of matter waves, creating an interference between the different contributions. In Feynman's formulation, the surface could be understood as the values of x at a certain time t, and the delay in the phase would be given by the action that would be needed in order to get from the surface to the point we want to get, as in classical mechanics. As Schrödinger equation depends only on first derivatives of time, there is no need to generalize the theory to fulfill Kirchoff's principle, which requires to know both the points of the wave and its first derivative.

As found in Eq. (12), we can understand temporal evolution as a recursive relation within a certain interval of time ϵ. If we apply that expression several times, we can determine ψ throughout the whole T. Let's now ask ourselves how can we recover the classical limit using this information. As Dirac pointed out in the 1930s [9], the classical limit $\hbar \to 0$ will be determined by the values that contribute the most to the integral. In our case, the region with the highest contribution will be the one that has a slowly varying argument in the exponential as x_i is modified. As the exponent depends on the action, this will hold whenever $\frac{\partial S}{\partial x_i} = 0$, which is the classical trajectory determined by Hamilton's principle of least action.

MATRIX ELEMENTS

Let's now compute the matrix elements of the different operators that we are interested in physics. First of all, note that the computation of matrix elements at different times is very natural, thanks to the formulation that we have been working with. In general, for a given operator $\hat{O}$, we can express its matrix element w.r.t. some given states ψ at time t' and χ at time t'' as:

$$\langle\chi_{t''}|\hat{O}|\psi_{t'}\rangle_S = \lim_{\epsilon\to 0}\int\cdots\int \chi^*(x'', t'')\hat{O}(x_0, \ldots, x_j)$$
$$\times \exp\left[\frac{i}{\hbar}\sum_{i=0}^{j-1} S(x_{i+1}, x_i)\right]\psi(x', t')\frac{\mathrm{d}x_0}{A}\cdots\frac{\mathrm{d}x_{j-1}}{A}\,\mathrm{d}x_j\,, \tag{18}$$

where $x' = x_0$ and $x'' = x_j$. In the article, Feynman exploits Eq. (18) in order to derive the transition elements of very well-known observables (e.g. position), as well as obtaining the canonical commutation relation.

HARMONIC OSCILLATOR

To put this formulation into practice, let's work out the most common system in Quantum Mechanics: the harmonic oscillator. Although Feynman discusses that we can eliminate the transversal modes in the equations of Quantum Electrodynamics via his new formulation, we will restrict ourselves to the harmonic oscillator by itself. To do that, we will suppose a general Lagrangian $L = a_0\dot{x}(t)^2 + a_1\dot{x}(t)x(t) + a_2x(t)^2 + a_3x(t) + a_4$, where the action will be given by its integration over time in a certain interval $T = [t_a, t_b]$. In order to find the energy spectrum, let's compute the function $K(\boldsymbol{b},\boldsymbol{a}) \equiv \langle \boldsymbol{x}_b, t_b|\boldsymbol{x}_a, t_a\rangle$, which is usually referred to as the *Feynman propagator* [10]. If we take into account that the classical trajectory fulfills the principle of least action, we can split $S[x]$ into two terms, $S[x] = S(\bar{x}) + F(t_a, t_b)$, where F is the function that takes into account the difference between the classical trajectory, $\bar{x}(t)$, and the trajectory we are taking into account, $x(t)$ (we will call this difference $y(t)$). Then, our Feynman propagator can be written as:

$$K(\boldsymbol{b},\boldsymbol{a}) = \exp\left(\frac{i}{\hbar}S[\bar{x}]\right)F(t_a,t_b); \tag{19}$$

$$F(t_a,t_b) = \int_{y_a=0}^{y_b=0}[\mathrm{d}y]\exp\left[\frac{i}{\hbar}\int_{t_a}^{t_b}(a_0\dot{y}^2 + a_1\dot{y}y + a_2y^2)\,\mathrm{d}t\right], \tag{20}$$

where the differential between square brackets indicates a product of all the differentials in the discretization (with their corresponding normalizations) and $y_a \equiv y(t_a)$, viceversa for t_b. The action of the classical trajectory, $S[\bar{x}]$, is easily found by using the typical Lagrangian for this problem. The following result is obtained:

$$S[\bar{x}] = \frac{m\omega}{2\sin(\omega(t_b-t_a))}\left[(x_a^2+x_b^2)\cos(\omega(t_b-t_a)) - 2x_ax_b\right]. \tag{21}$$

Now, we only need to find $F(t_a, t_b)$ in order to solve the problem. Using that the variation of the trajectory is null at the boundaries, we can write:

$$F(T) = \int\prod_{n=1}^{N-1}\frac{\mathrm{d}y_n}{A^N}\exp\left(\frac{im}{2\hbar\epsilon}\sum_{k=1}^{N}\left[(y_k-y_{k-1})^2 - \epsilon^2\omega^2y_k^2\right]\right), \tag{22}$$

where we have defined $T \equiv t_b - t_a$. Using usual techniques for discretization, one finally finds that:

$$F(T) = \sqrt{\frac{m\omega}{2\pi i\hbar\sin(\omega T)}}. \tag{23}$$

Putting all the results together, we can find the propagator for our harmonic oscillator:

$$K(\boldsymbol{b},\boldsymbol{a}) = \sqrt{\frac{m\omega}{\pi\hbar}}\sum_n C_n(x_a,x_b)e^{-i\omega T(n+1/2)}, \tag{24}$$

from where we can clearly identify the energy spectrum of the harmonic oscillator.

PERTURBATION THEORY

In terms of the Feynman propagator, the wave function $\Psi(x_k, t)$ after j time steps of size ϵ is defined by the expression below:

$$\Psi(x_{k+j}, t+j\epsilon) = \int K(x_{k+j}, x_k)\Psi(x_k,t)dx_k. \tag{25}$$

If we applied recursively the same time evolution as in Eq. (12), the exact form of the Feynman propagator would be well defined through Eq. (19) by setting $F(t_a, t_b) = 1$. However, if small time-dependent potential terms appear at each point in space, such that

$$V' = V_0(x(t),t) + \lambda V_I(x(t),t), \tag{26}$$

with $|\lambda| \ll 1$, we can assure that the expression of the propagator can be expanded in a Taylor series. This can be accomplished because λ is so small that $\int \lambda V_I(x(t),t)dt \ll \hbar$. Thus, in a first order aproximation in terms of λ, we obtain:

$$K(x_{k+j},x_k) = K_0(x_{k+j},x_k) + K_1(x_{k+j},x_k), \tag{27}$$

where $K_0(x_{k+j}, x_k)$ is the Feynman propagator without perturbations and the first order term satisfies the following expression:

$$K_I(x_{k+j},x_k) = \frac{-i\lambda}{\hbar}\times$$
$$\times\int_t^{t+j\epsilon}\mathrm{d}t_c\int K_0(x_{k+j},x_c)V_I(x_c,t_c)K_0(x_c,x_j)\,\mathrm{d}x_c\,. \tag{28}$$

Finally, considering Eq. (27), we have an expression through Eq. (25) for the perturbed wave function after a time period of $j\epsilon$. This expression allows us to find the matrix elements of the transition from one known state at an initial time t to another state at a final time t' when the particle is subject to a small potential.

CONCLUSIONS

To summarize, we have presented and analyzed Feynman's formulation of Quantum Mechanics, which is based on a least action principle as in classical mechanics. As we have discussed, this assumption yields a very powerful theory, due to the fact that all the possible trajectories of the particles are taken into account. Thanks to the discretization of the space, we are able to consider these trajectories by means of an integral over infinitely many variables and compute the wave function as the contribution of the trajectories from the past. Here lies the importance of the Feynman's propagator function, which determines the contribution of the wave function at the immediately preceeding instant to the wave function at the current instant from all possible positions, as expressed in Eq. (25). Furthermore, the Schrödinger equation is obtained by direct application of Eq. (12), a Taylor expansion and the Cauchy's residues theorem. This suggests that this formulation is valid, as it provides us with the equation that governs the evolution of wave functions.

Nevertheless, there are some problems yet to be addressed. First of all, it is left to work with Lagrangians that are not quadratic in the velocities. However, this issue is not of significant concern, as all the experiments that corroborated Quantum Mechanics are related to Lagrangians with the mentioned feature. Besides that, this formulation cannot be used for infinite intervals of times, as the action would diverge, as well as for general systems of coordinates. Also, we have seen that it is very useful for working with the position operator, as the formulation is based on a discretized spatial integral of infinitely many variables, but nothing is said about other observables such as linear momentum.

Finally, in order to witness the equivalence of this reformulation of Quantum Mecanichs w.r.t the previous ones, we have derived both the harmonic oscillator and the expressions for perturbation theory, confirming we obtain the usual ones.

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